Data Reconciliation Checklist: SCADA, EAM, CMMS, and Invoices

Category: Templates | Model: VFIP-TPL-RECON-CHK-2025 | Date Range: 2025-10-15 to 2025-12-15

Purpose: Practical checklist and template for analysts to align scada_telemetry_events, eam_asset_contract_master, cmms_work_orders_dispatch, and settlement_invoices_line_items. Includes firmware incident artifacts, runbooks, SQL skeletons, expected tolerances, reconciliation sample tables, and settlement guidance for ERCOT.

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Executive Summary & Scope

This document is a comprehensive reconciliation checklist and template intended for analysts and operations teams tasked with aligning the following systems and datasets for the window 2025-10-15 to 2025-12-15:

- SCADA: scada_telemetry_events (telemetry, alarms, time_bucket aggregation)
- EAM: eam_asset_contract_master (asset registration, contract availability)
- CMMS: cmms_work_orders_dispatch (corrective and preventive work orders)
- Invoices: settlement_invoices_line_items (settlement lines, penalty/charge amounts)

> Context: Synthetic document pack tied to the XG-440 firmware incident (v3.14.2 → v3.15.0). Use this template for RAG-based retrieval, QA, and forensics relating to plant-level generation, availability and settlement variance investigations.

Prerequisites, Data Sources & Date Filters

Required Data Exposures (Gold)

- gold_asset_availability — hourly / time_bucket availability percentages per plant/inverter
- gold_daily_generation — daily measured and scheduled MWh by plant and inverter_model
- gold_revenue_variance — revenue and settlement variance lines, mapped to invoices

Operational Data Sources & Tables

Source	Canonical Table	Primary Fields
SCADA	scada_telemetry_events	time_bucket, plant_id, inverter_serial, inverter_model, measured_mwh, status_codes, alarm_code
EAM	eam_asset_contract_master	asset_id, plant_id, contract_id, availability_pct_target, maintenance_window
CMMS	cmms_work_orders_dispatch	work_order_id, plant_id, inverter_serial, start_time, end_time, work_type, status
Invoices	settlement_invoices_line_items	invoice_id, plant_id, time_bucket, amount_usd, line_type, penalty_code

Date Range Filters

All queries and reconciliations in this document use the analysis window: **2025-10-15 (inclusive)** to **2025-12-15 (inclusive)**. Apply timezone normalization to UTC for all time-bucket matching.

Recommended SQL filter snippet:

```
WHERE time_bucket >= TIMESTAMP '2025-10-15 00:00:00+00'
  AND time_bucket <= TIMESTAMP '2025-12-15 23:59:59+00'
```

Stepwise Reconciliation Procedure

Overview

The reconciliation follows a three-pass alignment: 1) Inventory & schema verification; 2) Time-series alignment and aggregation; 3) Financial reconciliation and variance investigation.

Detailed Steps

1. **Inventory & Schema Verification**
    1. Confirm presence of required tables and latest gold exposures.
    2. Verify field types and unit conventions (MWh vs kWh). Ensure measured_mwh is in MWh.
    3. Cross-check inverter_serial, inverter_model, and plant_id mapping between SCADA and EAM.
2. **Time-series Alignment and Aggregation**
    1. Normalize all timestamps to UTC and align to canonical time_bucket intervals (hourly default).
    2. Aggregate scada_telemetry_events into hourly time_bucket per inverter: SUM(measured_mwh) as measured_mwh.
    3. Construct scheduled_mwh from gold_daily_generation prorated per time_bucket using plant schedule profile.
    4. Compute availability_pct = (operational_minutes / scheduled_minutes) * 100 or derived from gold_asset_availability.

3. **Event & CMMS Crosswalk**
    1. Map SCADA alarm codes and trip codes (e.g., TRIP-31, TRIP-44) to CMMS work order templates.
    2. For each time_bucket with reduced generation > tolerance, search cmms_work_orders_dispatch for related work_order_id.
4. **Financial Reconciliation**
    1. Join time_bucket-level generation delta to settlement_invoices_line_items using plant_id and time_bucket.
    2. Recompute expected invoice_amount_usd = measured_mwh * contracted_rate_usd_per_mwh + imbalance_charges.
    3. Flag variance rows where |invoice_amount_usd - recomputed_amount| > allowed tolerance (see Tolerances section).
5. **Validation & Traceability**
    1. For every flagged variance, produce an evidence packet: SCADA events, CMMS work orders, firmware change events, and invoice lines.
    2. Log findings into eam_asset_contract_master audit fields or a reconciliation log table: reconciliation_run_id, analyst_id, conclusion, remediation_required.

SQL Query Skeletons (Gold Exposures)

Use these skeletons as starting points. Replace schema and table names to match your environment. All examples assume UTC alignment.

1) gold_asset_availability (hourly)

```
-- gold_asset_availability skeleton
SELECT
  time_bucket,
  plant_id,
  inverter_model,
  ROUND(AVG(availability_pct), 4) AS availability_pct
FROM gold_asset_availability
WHERE time_bucket BETWEEN TIMESTAMP '2025-10-15 00:00:00+00' AND TIMESTAMP '2025-
GROUP BY time_bucket, plant_id, inverter_model
ORDER BY time_bucket, plant_id;
```

2) gold_daily_generation (daily)

```
-- gold_daily_generation skeleton (daily)
SELECT
  DATE_TRUNC('day', time_bucket) AS date,
  plant_id,
  inverter_model,
  SUM(measured_mwh) AS measured_mwh,
  SUM(scheduled_mwh) AS scheduled_mwh
FROM gold_daily_generation
WHERE time_bucket BETWEEN TIMESTAMP '2025-10-15 00:00:00+00' AND TIMESTAMP '2025-
GROUP BY date, plant_id, inverter_model
ORDER BY date, plant_id;
```

3) gold_revenue_variance (settlement)

```
-- gold_revenue_variance skeleton
SELECT
  si.time_bucket,
  si.plant_id,
  si.invoice_id,
  si.line_type,
  si.amount_usd AS invoice_amount_usd,
  grv.expected_amount_usd,
  (si.amount_usd - grv.expected_amount_usd) AS variance_usd
FROM settlement_invoices_line_items si
LEFT JOIN gold_revenue_variance grv
  ON si.plant_id = grv.plant_id
  AND si.time_bucket = grv.time_bucket
WHERE si.time_bucket BETWEEN TIMESTAMP '2025-10-15 00:00:00+00' AND TIMESTAMP '20
ORDER BY si.time_bucket, si.plant_id;
```

4) Time-bucket reconciliation join example

```
-- join SCADA hourly, CMMS, and invoices
SELECT
  s.time_bucket,
  s.plant_id,
  s.inverter_model,
  s.measured_mwh,
  sg.scheduled_mwh,
  a.availability_pct,
  w.work_order_id,
  i.invoice_amount_usd
FROM (
  SELECT time_bucket, plant_id, inverter_model, SUM(measured_mwh) AS measured_mwh
  FROM scada_telemetry_events
  WHERE time_bucket BETWEEN TIMESTAMP '2025-10-15 00:00:00+00' AND TIMESTAMP '202
  GROUP BY time_bucket, plant_id, inverter_model
) s
LEFT JOIN gold_daily_generation sg
  ON s.plant_id = sg.plant_id AND s.time_bucket = sg.time_bucket
LEFT JOIN gold_asset_availability a
  ON s.plant_id = a.plant_id AND s.time_bucket = a.time_bucket AND s.inverter_mod
LEFT JOIN (
  SELECT work_order_id, plant_id, inverter_serial, start_time, end_time
  FROM cmms_work_orders_dispatch
  WHERE start_time BETWEEN TIMESTAMP '2025-10-15 00:00:00+00' AND TIMESTAMP '2025
) w
  ON s.plant_id = w.plant_id
  AND s.time_bucket BETWEEN w.start_time AND w.end_time
LEFT JOIN settlement_invoices_line_items i
  ON s.plant_id = i.plant_id AND s.time_bucket = i.time_bucket
ORDER BY s.time_bucket, s.plant_id;
```

Sample Reconciliation Tables & Expected Tolerances

Below is a representative reconciliation table layout used in evidence packets. Use these

columns in export CSVs and RAG indices.

Sample: hourly_reconciliation_results

time_bucket	plant_id	inverter_model	measured_mwh	scheduled_mwh	availability
2025-11-20 10:00:00+00	PLANT-A	XG-440	2.150	2.200	97.7
2025-12-08 15:00:00+00	PLANT-B	XG-440	0.020	2.100	2.0

Expected Tolerances

Check	Tolerance / Threshold	Action if exceeded
Measured vs Scheduled MWh (hourly)	Absolute difference <= 0.2 MWh or <= 5% (whichever larger)	Flag for investigation; check CMMS & firmware events
Availability pct deviation (daily)	Deviation from contract target > 3 percentage points	Open corrective work order; initiate root-cause review
Invoice variance	|variance_usd| > 100 USD or > 2% of invoice line	Recompute settlement, escalate to settlements team
Missing CMMS linkage	No work_order_id for outage windows > 30 minutes	Create placeholder work order, notify O&M supervisor

Vendor Firmware Release Notes: XG-440 (v3.14.2 → v3.15.0)

This section summarizes synthetic vendor release notes relevant to the XG-440 incident for use in forensic reconciliation. The notes include parameter changes, protective trip codes, and recommended rollback conditions.

Release v3.14.2 (baseline prior to rollout)

- Stable baseline. Protective algorithm: dynamic_rst_enabled = true.
- Trip codes documented: TRIP-20 (overtemp), TRIP-31 (dc_isolation_fault).
- Known issues logged: occasional spurious TRIP-31 during high irradiance with rapid irradiance changes.

Release v3.14.9 (interim)

- Adjustments:
    - pv_input_filter_coeff reduced from 0.85 → 0.75 to smooth input spikes.
    - Trip sensitivity window for TRIP-31 tightened: trip_window_ms from 350ms → 250ms.
- Note: Regression risk for false DC isolation faults during irradiance ramp events.

Release v3.15.0 (rollout candidate)

- Major changes:

    - protective_trip_logic refactor — new prioritization: TRIP-44 (anti-islanding oversensitivity) introduced.
    - added parameter: trip_debounce_count (default 3).
    - voltage_sag_detection: enabled with threshold sag_v_pct = 15% under 50ms events.

- New protective trip codes:

Code	Description	Typical Symptom	Rollback Condition
TRIP-31	DC Isolation Fault (reworked logic)	Instantaneous inverter stops producing; alarms in SCADA: alarm_code='DC_ISO'	Repeat failures > 5 per plant per day after rollout → rollback to v3.14.2
TRIP-44	Anti-islanding oversensitivity (new)	Clusters of simultaneous trips across adjacent inverters during ramping	Widespread occurrence > 10% of fleet within 24h → rollback
TRIP-56	Voltage Sag Trip	Trips during short grid sags & low measured_mwh at same time_bucket	Rollback if incorrectly triggered during minor grid events

> - Recommended rollback conditions: If post-rollout monitoring detects cluster TRIP-31 or TRIP-44 events affecting >5% of inverters in the rollout group within 48 hours, initiate immediate rollback to v3.14.2 following the vendor rollback procedure.

Parameter Changes Summary

Parameter	v3.14.2	v3.15.0	Impact Notes
pv_input_filter_coeff	0.85	0.70	Smoother input; potential delay in transient detection
trip_window_ms	350	200	Tighter window increased sensitivity to short-duration faults
trip_debounce_count	--	3	Requires 3 triggers before trip; may reduce noise trips

Change Ticket Summaries & Rollout Details

Vendor portal change tickets related to the rollout and hotfixes. Include scope by plant and inverter serial ranges.

Rollout: 2025-11-20 → 2025-11-21 (Ticket: VND-CHG-2025-1120)

- Scope: Plants: PLANT-A, PLANT-B, PLANT-C
- Inverter serial ranges:

    - PLANT-A: XG440-1000 → XG440-1224
    - PLANT-B: XG440-2000 → XG440-2048
    - PLANT-C: XG440-3000 → XG440-3176
- Change summary:
    1. Firmware v3.15.0 staged to these serial ranges, phased by sub-bay groups over 24 hours.
    2. Monitoring window: 2025-11-20 00:00:00 to 2025-11-22 00:00:00 UTC.
    3. Rollback plan: immediate rollback to v3.14.2 for any cluster TRIP-31 or TRIP-44 occurrence as defined in release notes.

Hotfix: 2025-12-08 (Ticket: VND-HFX-2025-1208)

- Scope: Emergency hotfix applied to PLANT-B inverters serial XG440-2000 → XG440-2024 to address TRIP-31 frequency spikes.
- Hotfix details:
    - Parameter adjustment: trip_debounce_count temporarily increased to 5 for the hotfix.
    - Deployment: rolling update between 2025-12-08 02:00:00 and 2025-12-08 06:00:00 UTC.
- Observed: Post-hotfix, several PLANT-B inverters reported reduced tripping but anomalous low availability windows between 2025-12-08 15:00 and 2025-12-08 17:00 UTC (see O&M incident).

Change Log Records (example fields)

ticket_id	action	start_time	end_time	affected_serials
VND-CHG-2025-1120	rollout v3.15.0	2025-11-20 01:00:00+00	2025-11-21 23:00:00+00	XG440-1000..XG440-3176
VND-HFX-2025-1208	hotfix param change	2025-12-08 02:00:00+00	2025-12-08 06:00:00+00	XG440-2000..XG440-2024

O&M Incident Report & Postmortem: XG-440 Firmware Event

This O&M report is a synthetic, structured incident log for the XG-440 event window. Use for timeline correlation with SCADA, CMMS and settlements.

Incident Summary

Between 2025-11-20 and 2025-12-10 multiple plants reported elevated TRIP-31 and TRIP-44 occurrences following vendor firmware rollout v3.15.0. PLANT-B experienced widespread curtailment and low availability leading to settlement variances and imbalance charges.

Day-by-Day Timeline (Key Events)

Date (UTC)	Event	Impact	Action

2025-11-20 03:10	Rollout start (PLANT-A, PLANT-B)	Monitoring initiated	Telemetry alerting enabled; compare baseline
2025-11-20 10:45	Cluster TRIP-31 observed PLANT-A	Intermittent plant-level curtailment	Vendor notified; leverage change ticket
2025-11-21 04:00	Rollout paused	Containment	Rollback to v3.14.2 for subset of serials
2025-12-08 02:00	Hotfix deployment PLANT-B	Applied trip_debounce_count=5	Monitor for extended low availability
2025-12-08 15:00	Extended curtailment window PLANT-B	Many inverters reporting low measured_mwh	Open CMMS work orders; investigations
2025-12-09 09:00	Settlement team flagged invoice variances	Potential imbalance charges	Start reconciliation using this template
2025-12-10 18:00	Vendor supplied v3.15.0 patch v3.15.1 candidate	Test candidate with subset	Do not deploy fleetwide until tests complete

Curtailment Window Description (Example)

PLANT-B curtailment event on 2025-12-08 between 15:00–17:00 UTC. Characteristics:

- Time buckets with measured_mwh extremely low (<0.05 MWh for inverters normally producing 2.0–2.5 MWh).
- SCADA alarms: repeated TRIP-31 and TRIP-44 codes.
- CMMS: Multiple work orders auto-generated with template template_code='XG-FW-EMERG'.
- Settlement: invoice lines for those time_buckets contained imbalance charge line_type='IMB' with penalty_code='ERCOT-IMB-STD'.

Validation Steps Conducted

1. Correlate SCADA alarms to firmware change tickets and timestamps.
2. Verify CMMS work orders existence and closure status for the curtailment window.
3. Recompute expected generation using pre-rollout profiles and compare to measured results.
4. Calculate incremental imbalance charges and map to invoice lines for evidence.

Postmortem Findings (Key)

- Root Cause: sensitivity regression in protective_trip_logic (v3.15.0) caused spurious TRIP-31/TRIP-44 on rapid irradiance transitions.
- Contributing Factors: aggressive trip_window_ms reduction + pv_input_filter_coeff change.
- Mitigations: Vendor recommended immediate param rollback for affected serials

and patch v3.15.1 to adjust detection thresholds.

Dispatch Runbooks: Mapping SCADA Alarms to CMMS Work Orders

Standardized runbook entries ensure consistent corrective action creation and faster reconciliation.

Runbook: TRIP-31 (DC Isolation Fault)

1. SCADA Alarm: alarm_code = 'DC_ISO' or protective_code = 'TRIP-31'.
2. Automated CMMS Template: template_code = 'XG-DC-ISO-CHK'
3. Work Order Fields:
    - work_order_type = CORRECTIVE
    - priority = HIGH
    - assigned_team = EAM-ELECTRICAL
    - notes: include firmware_version, inverter_serial, alarm_history snapshot
4. Triage Steps:
    1. Confirm inverter is offline via scada_telemetry_events status_codes.
    2. Check firmware_version field in eam_asset_contract_master for recent changes.
    3. If multiple adjacent inverters show similar alarms, escalate to vendor and consider immediate rollback action via change ticket.

Runbook: TRIP-44 (Anti-Islanding Oversensitivity)

1. SCADA Alarm: alarm_code = 'ANTI_ISLAND' or protective_code = 'TRIP-44'.
2. CMMS Template: 'XG-ANTI-ISL-INVEST'
3. Investigation Steps:
    1. Gather grid event logs (grid_operator_feed) for correlation.
    2. Check if trip events cluster by bay or serial range; if yes, add vendor change ticket link in work order.
    3. Record measured_mwh before and after trip for settlement evidence packet.

CMMS Work Order Template Example (JSON-like fields)

```
{
  "template_code": "XG-FW-EMERG",
  "work_type": "CORRECTIVE",
  "priority": "HIGH",
  "required_fields": ["inverter_serial", "firmware_version", "alarm_code", "time_
  "escalation": {
    "thresholds": {
      "cluster_count": 5,
      "duration_minutes": 30
    },
    "notify": ["vendor_support@vendor.example", "operations@plant.example"]
  }
}
```

Settlement and Imbalance Charge Guidance (ERCOT)

This section outlines how imbalance charges are computed in ERCOT-style settlements and which data fields to reconcile.

Key Concepts

- Net Energy Imbalance = Scheduled MWh - Measured MWh for each settlement interval.
- Imbalance Charge = Net Energy Imbalance * RTSP (Real-Time Settlement Point) rate or applicable penalty rate.
- Relevant invoice line types: ENERGY, IMB (imbalance), ADJ (adjustments), PEN (penalty)

Essential Fields to Reconcile

Field	Description	Source Table
time_bucket	Settlement interval timestamp (UTC)	All
scheduled_mwh	ISO or contracted schedule for interval	gold_daily_generation
measured_mwh	SCADA measured energy	scada_telemetry_events / gold_daily_generation
rate_usd_per_mwh	Applicable RTSP or contracted rate	contract_rates / eam_asset_contract_master
invoice_amount_usd	Amount billed in invoice	settlement_invoices_line_items
penalty_code	Reason code for imbalance or special penalty	settlement_invoices_line_items

Reconciliation Procedure (Numeric Example)

Example metrics for a single time_bucket (2025-12-08 15:00:00+00)

Metric	Value
scheduled_mwh	2.100
measured_mwh	0.020
net_imbalance_mwh	2.080
RTSP (rate_usd_per_mwh)	200 USD/MWh
calculated_imbalance_charge	2.080 * 200 = 416.00 USD
invoice_line_amount (IMB)	430.00 USD
variance_usd	14.00 USD

Common Settlement Reconciliation Issues

- Time-bucket misalignment due to timezone or DST errors — ensure UTC normalization.
- Different rounding conventions between SCADA aggregation and settlements — use consistent rounding policy (two decimal places for USD, three for MWh when needed).
- Contract rate mismatches — validate eam_asset_contract_master for active contract_id and rates effective during the interval.
- Penalty codes mapping — verify penalty_code semantics with settlements team (e.g., 'ERCOT-IMB-STD' vs 'ERCOT-PEN-LATE').

Verification, Troubleshooting, and Escalation

Verification Checklist

1. Confirm UTC normalization and time_bucket matching across all datasets.
2. Verify inverter_serial and inverter_model mapping exists in eam_asset_contract_master for all serials appearing in SCADA events.
3. Check firmware_version field for each inverter involved in variance events against vendor change logs.
4. Ensure CMMS work orders reference the time_bucket windows and include vendor ticket IDs if related to firmware changes.
5. Validate computed imbalance charges against invoice lines with a tolerance threshold.

Troubleshooting Flow (Textual Flowchart)

Start → Are measured_mwh values abnormal relative to schedule? → Yes → Any CMMS work order in time window? → Yes → Use CMMS notes & firmware change tickets to explain variance; tag as *explained*. → No → Check SCADA alarms for TRIP codes → If TRIP-31/TRIP-44 present, check firmware version → If firmware change within 48h, link to vendor ticket and escalate to vendor. → If no alarms, check telemetry integrity (packet loss, GPS sync) → If telemetry integrity issue, open Data Quality ticket and hold settlements adjustment.

Escalation Matrix

Issue	Primary Contact	Escalation Contact	SLA
Spurious TRIP-31 across multiple plants	Operations Lead (ops-lead@company.example)	Vendor Support (vendor_support@vendor.example)	2 hours
Settlement variance > 5k USD	Settlements Manager (settlements@company.example)	Finance Director	4 hours
Missing critical telemetry	SCADA Engineer	Network Team	24 hours

Appendix: Field Definitions, Codes, and Change Log

Field Definitions

Field	Type	Description
time_bucket	TIMESTAMP (UTC)	Canonical interval start timestamp
plant_id	STRING	Plant identifier (e.g., PLANT-A)
inverter_serial	STRING	Unique inverter serial number
inverter_model	STRING	Model code (e.g., XG-440)
measured_mwh	DECIMAL	Measured energy in MWh for the interval
scheduled_mwh	DECIMAL	Scheduled energy in MWh for the interval
availability_pct	DECIMAL	Availability percentage (0–100)
work_order_id	STRING	CMMS work order identifier
invoice_amount_usd	DECIMAL	Amount on settlement invoice line
penalty_code	STRING	Settlement penalty identifier

Protective Trip Codes (Reference)

Code	Meaning	Typical SCADA Alarm Field
TRIP-20	Overtemperature	ALARM_OVER_TEMP
TRIP-31	DC Isolation Fault	ALARM_DC_ISO
TRIP-44	Anti-islanding Oversensitivity	ALARM_ANTI_ISL
TRIP-56	Voltage Sag Trip	ALARM_VOLT_SAG

Example Reconciliation Run Log Record

```
-- reconciliation_log example insert
INSERT INTO reconciliation_log (
  reconciliation_run_id,
  run_start_ts,
  run_end_ts,
  analyst_id,
  rows_processed,
  flagged_rows,
  notes
) VALUES (
  'RUN-20251210-001',
  TIMESTAMP '2025-12-10 08:00:00+00',
  TIMESTAMP '2025-12-10 09:30:00+00',
  'analyst.jane',
```

```
  12480,
  32,
  'Investigated PLANT-B curtailment 2025-12-08; evidence packet attached; vendor
);
```

Change Log for this Template

Version	Date	Author	Notes
VFIP-TPL-RECON-CHK-2025 v1.0	2025-12-11	Ops Analytics	Initial template including XG-440 incident artifacts and SQL skeletons

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Prepared by: Operations Analytics — Template Model VFIP-TPL-RECON-CHK-2025. Use this template for internal reconciliation and RAG system indexing. Retain vendor ticket references and CMMS links in evidence exports. For legal or regulatory use, consult settlements and legal teams before releasing excerpts.